Unit: 6 Unit 8: Quadratics
Topic: Vertex Form and Intercept Form

Name: ________________
Date: ____________

Foundation (Jalapeno)

Q1. The quadratic function $f(x) = (x-3)^2 + 2$ is in $\boxed{vertexform}$.
(A) standard form
(B) vertex form
(C) intercept form
(D) factored form
(E) none of these

Q2. The vertex of the quadratic $f(x) = x^2 - 6x + 8$ is $(h, k) = (\boxed{3}, -1)$.
(A) (1, -1)
(B) (3, -1)
(C) (3, 2)
(D) (-3, 2)
(E) (0, 0)

Q3. For the quadratic function $f(x) = a(x-h)^2 + k$, h and k represent the $\boxed{x}$- and y-coordinates of the vertex.
(A) minimum and maximum
(B) maximum and minimum
(C) x- and y-coordinates of the vertex
(D) slope and y-intercept
(E) none of these

Q4. The x-intercepts of the quadratic function $f(x) = x^2 - 4x - 5$ are $(x, y) = (\boxed{-1}, 0)$ and $(5, 0)$.
(A) (-1, 0) and (1, 0)
(B) (-5, 0) and (1, 0)
(C) (-1, 0) and (5, 0)
(D) (1, 0) and (5, 0)
(E) (-5, 0) and (-1, 0)

Q5. The quadratic $f(x) = (x+2)(x-4)$ is in $\boxed{factoredform}$, which can help find the x-intercepts.
(A) standard form
(B) vertex form
(C) intercept form
(D) factored form
(E) none of these

Q6. A basketball player throws a ball with an initial velocity of 25 feet per second. The height h of the ball after t seconds can be modeled by the quadratic equation $h(t) = -16t^2 + 25t + 5$. The vertex form of this equation is $h(t) = -(t - \boxed{0.78125})^2 + 20.15625$.
(A) 0.625
(B) 0.78125
(C) 1.25
(D) 1.5625
(E) 1.875

Q7. For a tennis serve, the trajectory of the ball can be modeled by $h(x) = 0.2(x-10)^2 - 4$, where $h(x)$ is the height of the ball in feet and x is the distance in feet. The y-intercept of the graph is $(0, \boxed{-4})$.
(A) (0, 0)
(B) (0, -4)
(C) (10, 0)
(D) (0, 4)
(E) (0, -10)

Q8. The quadratic function $f(x) = (x-2)(x+1)$ can be expressed in intercept form as $f(x) = (x - \boxed{2})(x - (-1))$.
(A) (x - 1)(x + 1)
(B) (x - 2)(x + 1)
(C) (x - (-1))(x + 2)
(D) (x - 1)(x - 2)
(E) (x + 1)(x - 2)

Open Ended Questions (FRQ)

Q9. A baseball is hit with an initial velocity of 30 feet per second at an angle of 45^circ. The height h of the ball after t seconds can be modeled by $h(t) = -16t^2 + 30t$. Rewrite this equation in vertex form and determine the vertex. Provide your answer in the format (h, k), where h and k are the x- and y-coordinates of the vertex.

Q10. The path of a golf ball is modeled by the quadratic function $f(x) = 0.2(x-50)^2 + 10$, where $f(x)$ is the height of the ball in feet and x is the distance in feet. Determine the x-intercepts and y-intercept of this quadratic function. Show all your work and calculate your answers to the nearest tenth.

Chef's Tips

- Tip 1: Ensure students understand the definitions of vertex and intercept forms before attempting these questions.
- Tip 2: Students should be given extra practice converting between standard form, vertex form, and intercept form.
- Tip 3: Real-world sports examples will help students connect the mathematical concepts to their everyday lives.